

Fundamentals of Machine learning for and with engineering applications

Enrico Riccardi¹

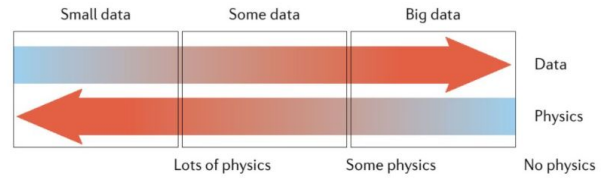
Department of Mathematics and Physics, University of Stavanger (UiS).¹

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Data vs Physics



Uncertainty

- Def 1: Not knowing if an event is true or false. (Useful)
- Def 2: Things that cannot be measured. (Not useful)

Probability is how Uncertainty is quantified!

- Clarity test
- Assign a number between 0 and 1 to our degree of belief
- Error definition

Sentence also good for fortune cookies

Uncertainty is the only certainty

Uncertainty and Probability

Random quotes

- Probability: there is not science more worthy in our contemplations nor a more useful one for admission to our system of public education
- The theory of probabilities is at the bottom of nothing but common sense reduced to calculus.

What is Statistics

Clarity test. Beer drinker?

Rain in Stavanger?

Descriptive and Predictive statistics

Estimation

- Process of obtaining the best value or range of a property in an unsampled location
- Local accuracy takes precedence over global spatial variability
- Not appropriate for forecasting

Inference

- Predict unseen samples given assumptions about the population
- Test with a pre-trained model (ML definition)
- Generality versus Accuracy

Variables and Features, Labels and Instances

Population

Exhaustive, finite list of properties of interest over area of interest.

Generally the entire population is not accessible

Samples/experiments/instances

The set of values and location that have been measured.

How many experiments are needed?

Sampling distribution -(IID ?)

identical and independent distribution The set of values and location that have been measured.

Features

The values to be measured for each sample/experiment/instance.

How many features are needed?

Variables and Features, Labels and Instances

Predictors = input variables, $X_1, \dots, X_M$

Response = output variables

Error

Deviation from ... exact value (or expected value, mean value, trend...?)

Errors without definitions are just numbers.

Error

Values, without error, are just number!!

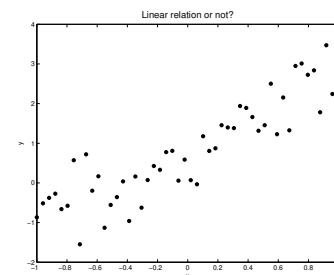
Predictor and Response Features

Given a model $Y = f(X_1, \dots, X_M) + e$

!Here and error! But is it even an error?

Finding a suitable model

Soft modeling is in most cases based on **multivariate statistical methods**. Many of these methods may be viewed as sophisticated ways of performing curve fitting to data.



What would be the best model?

- Straight?: $y(x) = ax + b$
- Parabolic?: $y(x) = ax^2 + bx + c$
- Trigonometric?: $y(x) = a\sin(x) + b\cos(x)$

Uncertainty Modeling

Given a model, Generate multiple simulation to represent uncertainty

- Realizations: for the same input parameters, different random numbers.
- Scenarios: different input parameters.

Sampling representative.

Random sampling

Each item of the population has an equal chance of being chosen.

- Very expensive
- Mostly not interesting
- Gives some global properties

Bias sampling

Selection of data is (arbitrarily) distorted

- Sample probability bias has to be corrected for
- Might not capture the global picture
- It might distort the system under study -> false results

Cognitive biases

- anchoring: The first bits are over-considered
- availability: over-estimating the importance of info
- bandwagon: P increases with the number of people holding a belief
- blind spot: not seen biases
- choice supporting: commitment/decision dependent
- clustering illusion: seeing patterns in random events
- confirmation bias
- conservatism bias
- Recency bias
- Supervision bias
- Many many more!

Bias DO NOT cancel out! They sum up (or multiply?)

Simulations

Process of obtaining one or more values of a property

- Improved Global accuracy
- Better property distributions

Why simulations then?

- We need to capture the full distribution of properties, extremes matter!
- We need more realistic models.

Why not?

- High dimensionality level
- Computationally expensive
- Convergence limitations
- Constitutive equations need to be rather accurate.

Data properties

DATA



SORTED



ARRANGED



PRESENTED VISUALLY



EXPLAINED WITH A STORY



Types of data

- Categorical / Nominal (classes)
- Categorical / Ordinal
- Continuous / Interval (e.g. Celsius)
- Continuous / Ratio
- Discrete: binned/grouped data
- Hard data: direct measurements
- Soft data: indirect measurements, very uncertain
- Primary data: variable(s) of interest
- Secondary data: descriptors
- Collective variables
- Latent variables

Representation

A representation should **capture** the nature of the subject being studied.

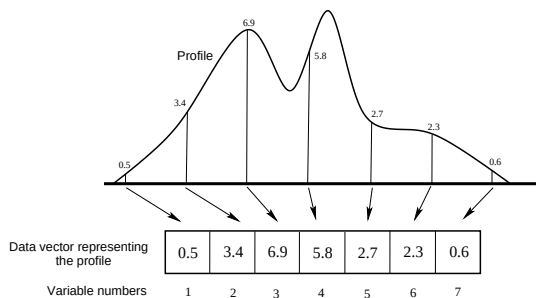
Example: If you want to evaluate the 3D structure of a wind turbine, a set of descriptors can be:

- 1 Blade length
- 2 Turbine height
- 3 Geographical position
- 4 Output power
- 5 Wind direction

which are two decimal numbers, a 2d tuple, a 1D time series and a 2D time series (or 3D even).

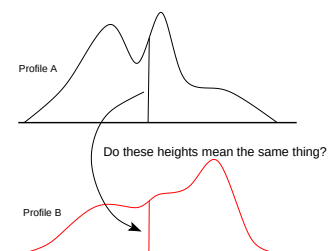
Sampling point representation (SPR)

- An intuitive way to represent curves and spectra is the **sampling point representation**.
- We sample at regular intervals where each sample point is represented by a variable



Sampling point representation (SPR)

- SPR is useful until point i in a curve has the same meaning of the point i in another curve.



- Which parts of the profiles or shapes are comparable, i.e. have the same meaning?

Comparability

Same meaning **representations** for different objects (inputs).

Discussion point!

How do we compare two wind turbines accounting for the 5 variables previously introduced?

Good data and bad data

- All starts from data: what are data-properties?
- Are there such things as good data and bad data?

Life lesson (or exam question, same thing :))

- Data **DO NOT** always have value.

- TRASH in TRASH out

Data structures

Given a representation, it is then needed to decide on a suitable **data structure** for the problem.

Definition

A data structure is a way of storing and organising data in a computer so that it can be used effectively.

Typical data structures used in data analysis are:

- Data points
- Arrays (vectors, matrices, N-mode (way) arrays)
- Graphs (trees)
- Databases

Workflow

Data has to be prepared with these steps in mind

- 1 Plan experiments: Use experimental design to set up experiments in a *systematic* way
- 2 Pre-processing: Is there systematic variation in the data which should be removed? Can cross-checking/validation procedures be designed?
- 3 Examine the data: Look at data (tables and plots). Strange behaviours? Smooth behaviour? WARNING!
- 4 Define desired model outcomes (speed, accuracy, false positive/negatives rate)
- 5 Estimate and validate model: What do the results tell us? Is the generated model general (valid for future sampling)?
- 6 Apply model to unknown samples

Data dimensionality

1 D

logs

2 D: maps

Quite limited but great for visualization

3 D

3d maps, seismic cubes. More informative, mostly ok in digital formats.

4 D

Trajectories

x D

Data realm

Spatial and Temporal Data

Statistics is collecting, organising, and interpreting data

Spatial and temporal statistics is a branch of applied statistics that emphasises:

- 1 the geo context of the data
- 2 the spatial and time dependent relationship between data
- 3 the different relative value and precision of the data.

Actual data

The data matrix is an extremely common data structure.

$$X = \begin{bmatrix} 95 & 89 & 82 \\ 23 & 76 & 44 \\ 61 & 46 & 62 \\ 49 & 2 & 79 \end{bmatrix}$$

In python these can be saved as

- lists (vanilla python)
- numpy arrays
- pandas dataframes

Nomenclature

There are different conventions. Commonly we will construct data matrix such that:

- Rows are called instances, objects or samples.
- Columns are called features, variables.

One can think of each row to be an experiment, and the rows its properties. Each row (experiment, object, sample, ...) is thus a list of values, one for property.

Note

Mathematically speaking, this is just a notation. As long as one keeps track and is consistent, columns can be used as rows and vice versa.

A quick example

Environmental measurements of rivers. Features (properties):

- pH
- Temperature
- Concentration of pollutants
- Flow rate
- water speed

The instances/experiments/observations/sample:

- Po
- Danube
- Rio delle Amazzoni
- Sjoa
- Atna

Data into Machine Learning algorithms: normalisation

Different measurements can lead to different data distribution and numerical values.

When fed to numerical recipes, data is just a set of numbers.

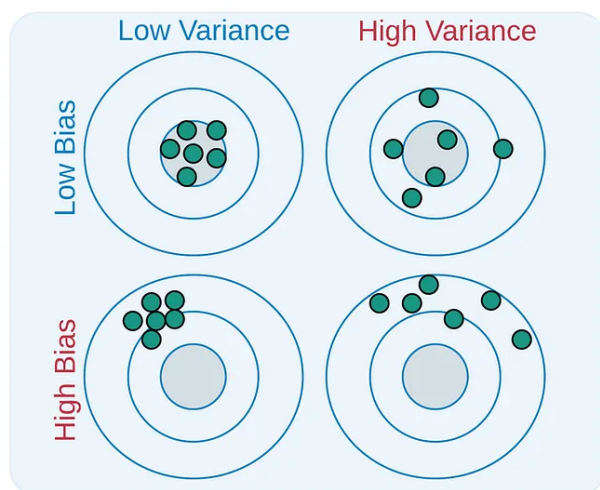
For faster computations, data can be **normalized**: i.e. rescaled to the same numerical interval.

Normalisation

The procedure simplifies the numerical recipes work, but it also loses information

As it is not possible to complete sample any distribution, data might be disproportionately handled depending upon data collection (i.e. this can be an error source).

Bias-Variance



Trusting the data

- Clarity test

project

- Measured or asserted Probability ?

project

Risk

- Risk = probability * consequences
- Value of information

project

- Risk attitude

project

Risk: vibe coding

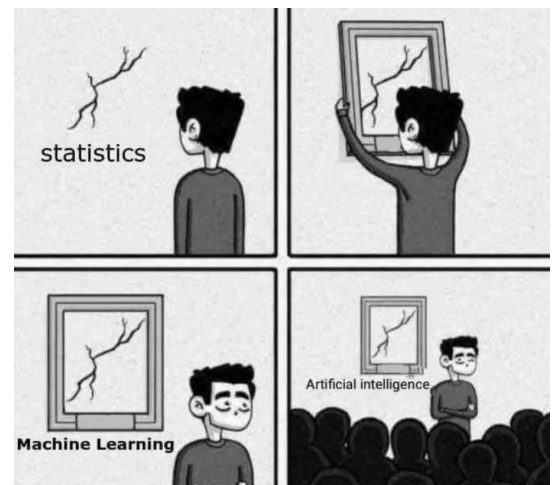
Vibe coding, what possibly can go wrong?



Risk: bow tie project



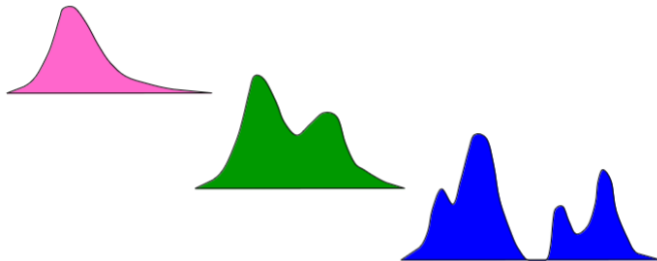
A bias recap



Probability distribution

A metric is needed
Amount of uncertainty

Number of Modes: unimodal, bimodal, polymodal



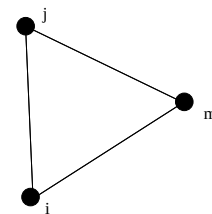
Metrics

Triangle inequality

Considering a vectors in an N-dimensional space, to be a **distance** it must satisfy the **triangle inequality**:

$$d_{ij} + d_{im} \geq d_{jm}$$

If also $d_{jj} = 0$, if $d_{ij} = d_{ji} = 0$, then we call it a *metric*.



Metrics example

Common metrics:

- Counting
- Euclidean

$$d_{ij}^{(E)} = \left[\sum_{k=1}^N (x_{ik} - x_{jk})^2 \right]^{\frac{1}{2}}$$

Categorical variables do not necessarily satisfy these relations.

Task assignment

Code a discrete probability distribution in Python

Calculate the Mean and Standard Deviation

How to get an experiment out of this distribution?

How to relate a distribution to information?

First, the origins:

- Originator: Claude Shannon (1940)
- His initial work has been done on signal transmission.
- It uses Entropy as key measurement of information uncertainty.

How Entropy can alter a decision?

It is an interface between data and decisions.

A question to sum up the idea

Does more data bring value?

It has permitted the advances of several fields:

cryptography, neurobiology, signal processing, linguistics, bioinformatics, statistical physics, black holes, quantum computing, information retrieval, intelligence gathering, plagiarism detection, pattern recognition, anomaly detection, etc

zip files, phones, internet!

Entropy of an information source

$$H_X = -\sum_i (p_i) \log(p_i)$$

H_X of a discrete random variable X is a measure of the amount of uncertainty associated with the value of X when only its distribution is known

What is p_i ?

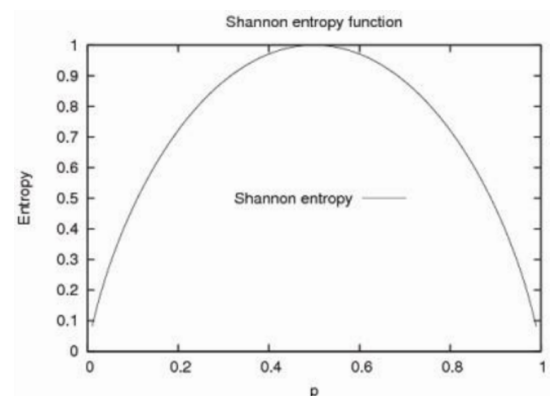
It is a numerical descriptions of how likely an event is to occur

Do not mix the concepts!

Assigned probability and computed probability are different

Shannon's entropy shape

$$H_X = -\sum_i (p_i) \log(p_i)$$



Task assignment [2]

Code a discrete probability distribution in Python

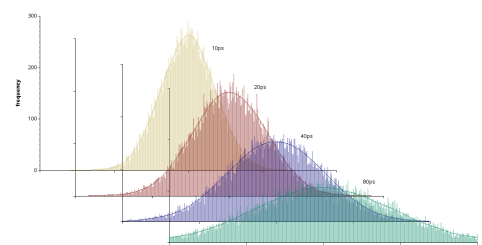
Calculate the Mean and Standard Deviation

How to get an experiment out of this distribution?

Calculate Shannon's Entropy as a function of the number of experiments

Comparing information

A distribution can be the sum of multiple distributions



Kullback–Leibler divergence (information gain)	Task assignment [3]
How to compare multiple information sources? $D_{KL} = \sum_i (p_i) \log\left(\frac{p_i}{q_i}\right)$ • It is a divergence, as it is asymmetric • It is NOT a metric	Compare the initial discrete probability distribution with the approximate ones computed as function of the number of experiments. Calculate Kullback–Leibler divergence for different number of experiments Find the distribution that minimise K-L divergence